

Hikmah
HACKERS

Week 1: Hello, Python!

Getting set up and writing our very first program

DARUL ULOOM MICHIGAN · FRIDAYS

TODAY

Our plan for 40 minutes

Time	What we do
5 min	Welcome and class rules
8 min	What is programming? Why Python?
10 min	Open the Python editor on your Chromebook
12 min	Write and run your first program
5 min	How to save, and this week’s mini-project

THE BIG IDEA

Programming means giving a computer exact instructions, in the right order.

1

Get two slices of bread

2

Put the filling on the bread

3

Put the second slice on top

Remember the sandwich question from the pretest? Computers follow steps exactly, one by one.

OUR LANGUAGE

Why Python?

It reads almost like English, and real programmers use it every day.

Games

Guessing games, quizzes, even simple arcade games

Websites and apps

The code that runs behind many sites you use

AI and science

Robots, space missions, and artificial intelligence

SETUP

Our tools this year

NOW

A Python editor in the browser

Nothing to install, no login

LATER

VS Code and GitHub

The tools real programmers use

GOAL

Our own website

Lesson plans, assignment submissions will live here

Our Chromebooks can’t install apps, so we start in the browser. Same Python, just a different window.

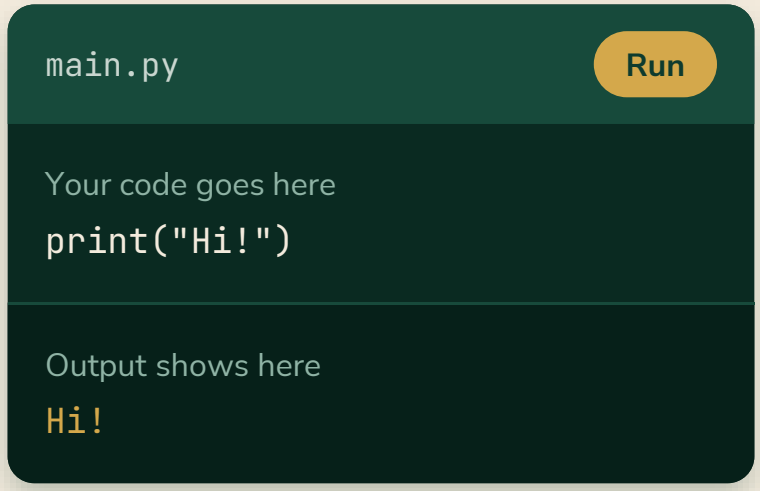
SETUP · 10 MIN

Open the Python editor

1. Open **Chrome** on your Chromebook
2. Type this into the address bar:

   ```
   programiz.com/python-programming/online-compiler
   ```

3. Delete any code already in the box
4. Find the **Run** button. You'll press it a lot!



Page blocked or not loading? Put your sticky note up and we'll switch to the backup.

CLASS RULES

How we help each other

Ask a partner first

Your neighbour might already know the answer.

Still stuck? Make a document of notes

Open a doc and write notes

Errors are normal

Every programmer gets errors. Fixing them is how you learn.

Bring sticky notes, one per student.

LET'S CODE · TYPE ALONG

Your first program

TYPE THIS, THEN PRESS RUN

```
print("Hi!")
```

YOU SHOULD SEE

```
Hi!
```

This is a canon event for every Python coder.

HOW IT WORKS

Three parts of a print line

```
print("Hello, World!")
```

`print`

The command. It means “show this on the screen.”

`( )`

Brackets hold what you want to print.

`" "`

Quotes mean “this is a string.” We will learn more about this later.

YOUR TURN · 5 MIN

Make Python print three lines about you

```
print("My name is Mohammad")
print("I am in 9th grade")
print("I love baat")
```

Finished early?

Add a fourth line. Then swap the order of two lines and press Run. What changed?

Use your own name, grade, and favourite food.

BUG HUNT

Three mistakes everyone makes

BROKEN	WHAT'S WRONG	FIXED
`print("Hi)`	Missing the closing quote	`print("Hi")`
`print("Hi"`	Missing the closing bracket	`print("Hi")`
`Print("Hi")`	Capital P. Python cares about capitals!	`print("Hi")`

Red text in the output is Python telling you what to fix. Read the last line first.

DON’T LOSE YOUR WORK

The editor doesn’t save. You do.

`Ctrl + A`

Click in your code, then select all of it

`Ctrl + C`

Copy it

`Ctrl + V`

Paste into a Google Doc called **Hikmah Hackers – My Code**

Use the same Google Doc every week. Put a heading for each week.

MINI-PROJECT · DUE NEXT FRIDAY

All About Me

Required

Write a program with **at least 5 print lines** about you:

- Your name and grade
- A hobby or sport you like
- Your favourite food
- Something you want to build with code

Bonus: draw with print

```
print(" /\ ")
print(" / \ ")
print(" | | ")
print(" |____| ")
```

Make a picture out of symbols: a house, a rocket, your initials…

Turn in: paste your code into Google Classroom (class code: [_____]). We'll show a few next Friday!

NEXT FRIDAY

Variables: how to remember things

```
name = "Mohammad"
print("Salaam, " + name)
```

Good work. You're officially coders. Soon you can be Iron-man. Assalamualaikum.